

Ethernet (T1/E1) because these have common characteristics like connection-oriented protocols, association with frame-level functioning and transfer of IP packets between adjacent nodes only.

So data-link layers (such as frame relay or ATM), layer-2 identifiers, such as data-link connection identifiers (DLCIs) in the case of frame-relay networks, or virtual path identifiers (VPIs)/virtual channel identifiers (VCIs) in case of ATM networks, or MAC headers in the case of PPP/Ethernet, can be used directly as labels. Packets are then forwarded based on their label value.

Labels are bound to an FEC as a result of some event or policy that indicates a need for such binding. These events can either be data-driven bindings or control-driven bindings. The latter is preferable because of their advanced scaling properties that can be used in MPLS. Policy of label binding is based on destination unicast routing, traffic engineering, multicast and quality of service.

Label merging and stacking

Label merging is the replacement of multiple incoming labels for a particular FEC with a single outgoing label (Fig. 4). It is achieved by stacking the label at LSR based on the instant of arrival of packets through incoming LSPs. It is organised as a last-in, first-out stack.

Protocols used in MPLS network

An MPLS network uses different types of protocols like open short path first (OSPF), border gateway protocol (BGP), protocol-independent multicast (PIM) and resource reservation protocol (RRP/RSVP).

OSPF is a routing protocol that multicasts a change in the routing table of a host to all other hosts within the boundary of a network. In MPLS network, this protocol is used as a label-distribution protocol between peers. This protocol is one among interior gateway protocols (IGPs).

BGP is also a routing protocol

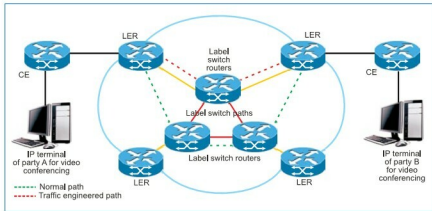


Fig. 5: Traffic engineered route in MPLS

that provides loop-free inter-domain routing between autonomous systems. An autonomous system is a set of routers that operate under the same administration. Here, an MPLS domain becomes an autonomous system. BGP is often run among VPN networks and MPLS networks.

PIM is used for multicast states label mapping. Some users may want to broadcast their messages to different users. This protocol supports the distribution of multicast labels. As a result, multiple LSPs are formed between single users to multi-users during broadcast period only.

RRP is not a routing protocol. It works in conjunction with other routing protocols to keep the quality of service within the MPLS cloud. It uses exchanging of labels pertaining to the services that require time management (online services like video conferencing, IP telephony and the like). RSVP provides the creation of tunnels in MPLS domain.

Traffic engineering in MPLS

Traffic engineering is essential for optimising utilisation of network resources. Network resources should not be wasted. At the same time, quality of service should be maintained for users.

In MPLS, layer 3 is overlaid with connection-oriented switching function of layer 2. By using this property, traffic-engineered dedicated paths for a different category of IP packets can be defined to maintain the quality of

service. Thus, an MPLS network is converted into a homogeneous network to handle heterogeneous traffic. These dedicated paths are known as traffic engineered tunnels (Fig. 5).

Different types of traffic engineered tunnels are created based on the quality of service of different users. Thus, these tunnels provide an alternate path that has been deliberately configured to be used as an alternative, unused path through the network in an effort to utilise network resources in an efficient way.

Conclusion

In an MPLS domain, routers do not analyse the entire IP address to select the best matching but only analyse the label (similar to the analysis of route code in circuit switching), which greatly reduces delay in routing data packets. Thus, the amount of per-packet processing required at each router in an IP based network is greatly reduced with the use of MPLS.

Moreover, label switch paths and labels are selected for routing by label switch routers according to the FEC of that IP packet (category and priority). Since MPLS is a connection-oriented transmission protocol, chances of packet loss are rare. Security is ensured because of label swapping.

MPLS supports new capabilities efficiently and with full integrity, which has ensured its popularity in areas like virtual private network (VPN), Intranet, VoIP and video conferencing. ●